

Solution of MGMT 670 Midterm Exam (Version A)

Fall 2024

Name (Print): _____

Student ID No.: _____

Section: _____ 1: MWF 2:50 – 4:20 pm
_____ 2: MWF 4:30 – 6:00 pm

EXAM Rules:

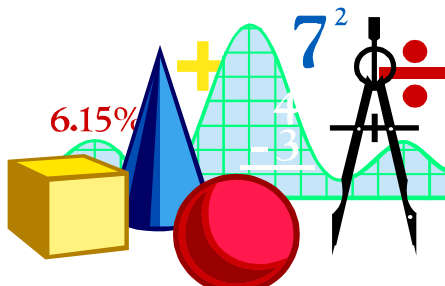
This is an open-book, open-notes exam. **No cellphone. No internet. No cheating!**

PART I. MULTIPLE CHOICES

1. Circle your answers on the exam.

PART II. WORK PROBLEMS

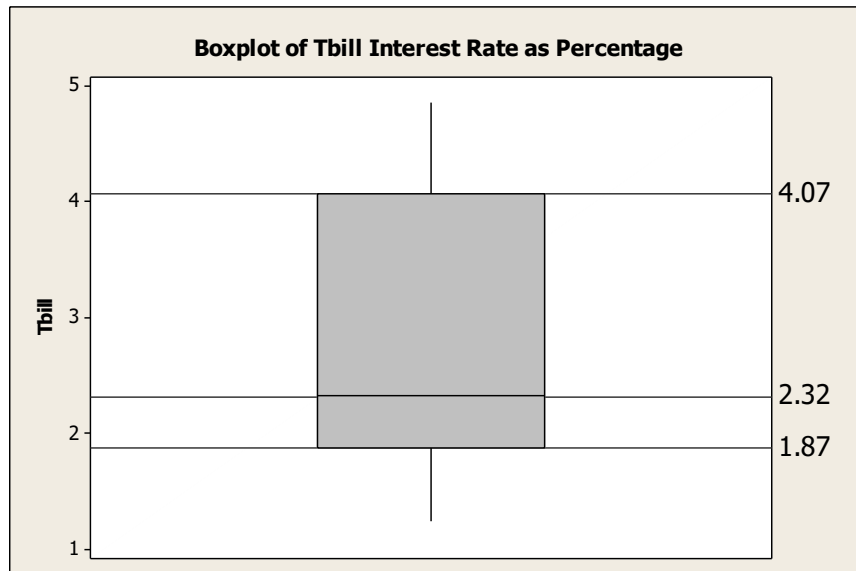
1. Give neat, clean, and well organized answers.
2. Show your work to support your answers.



Part I: Multiple Choices (10 Questions, 30 Points Total)

Identify the letter of the choice that **best** completes the statement or answers the question.

1. Which of the following is true about correlation coefficient?
 - a. Correlation coefficient is the proportion of points that do not lie on a straight line.
 - b. Correlation coefficient lies within the interval $[0, 1]$.
 - c. Correlation coefficient is independent of the change of scale of the data.
 - d. If the data lie on a straight line, the correlation coefficient is 1.
2. Annual sales of a chain restaurant at different locations do not quite look like a bell curve. The average is \$6 million and standard deviation is \$1.5 million. What can you say about the percentage of locations with sales between \$3 million and \$9 million?
 - a. No more than 50% of the restaurants are in this category
 - b. At least 75% of the restaurants are in this category
 - c. 95% of the restaurants are in this category
 - d. None of above is correct.
3. Treasury bills (also known as T-bills) are bonds issued by U.S. Department of the Treasury. You buy them at a discount from their face value, and they mature in a fixed period of time. For example, you might buy a \$1000 T-bill for \$990. When it matures, say six months later, you receive \$1000 – your \$990 and \$10 in interest. Interest is usually reported as a rate per year and would be 2.02% for the example setting. Rates are determined by an auction that is held every four weeks. The data set contains auction rates for T-bills from December 12, 1958 to October 3, 2008. A box plot has been drawn using MINITAB for your interpretation. Which of the following is true?



- a. The average interest rate for T-bills issued is less than 2.32%.
- b. Fifty percent of the T-bills issued had an interest rate between 2.32% and 4.07%
- c. All the interest rates in this data set are lower than 5%.
- d. None of the above

4. Suppose flights of Delta airlines depart on-time with a probability of 0.90. The airlines want to predict the delays among 12 flights that will depart tomorrow. Let X be the number of delayed flights among these 12 flights. Compare Q_1 and Q_2 that are defined as follows.

$$Q_1 = P(X \geq 1) ; Q_2 = P(X \leq 1)$$

What is the relationship between Q_1 and Q_2 ?

- a. $Q_1 > Q_2$
b. $Q_1 < Q_2$
c. $Q_1 + Q_2 = 1$
d. None of the above
5. Quality engineers at an electric company evaluated performances of inspectors in judging the quality of solder joints by comparing each inspector's classifications of a set of 150 joints with the consensus evaluation of a panel of experts. The results for a particular inspector are shown in the table. One of the 150 joints was selected at random. What is the probability that the inspector and the committee agreed on their judgement about the joint?

	Inspector found acceptable	Inspector found not acceptable
Committee found acceptable	101	10
Committee found not acceptable	20	19

- a. Less than 30%
b. Between 30% and 60%
c. Between 60% and 75%
d. More than 75%
6. Price scanners at the checkout register sometimes make mistakes. Suppose a scanner is wrong 4% of the time, independent of the price of the item. Further, assume that half the errors increase the original price by \$2 and half the errors reduce the original price by \$2. What would be the impact on the expected price of the items sold?
- a. The expected price will not change
b. The expected price will go up
c. The expected price will go down
d. Cannot be determined from the statement of the problem
7. A machine is used in a production process. From past data, it is known that 95% of the time the machine is set up correctly. Furthermore, it is known that if the machine is set up correctly, it produces 97% acceptable (non-defective) items. However, when it is set up incorrectly, it produces only 40% acceptable items. Given that the selected item is non-defective, what is the probability that the machine is set up correctly?
- a. 0.9871
b. 0.9787
c. 0.9415
d. 0.9215

8. Akiko is a manager at a sushi restaurant. She knows the importance of purchasing the right amount of salmon daily. While purchasing too much salmon results in wastage, purchasing too little can disappoint customers. Akiko has estimated that the daily consumption of salmon is normally distributed with a mean of 12 pounds and a standard deviation of 3.2 pounds. At least how much salmon should she buy so that it meets customer demand on 90% of the days? Round to 1 decimal.
- 17.3 pounds
 - 16.1 pounds
 - 14.6 pounds
 - 7.9 pounds
9. National Car Rental Systems Inc. Commissioned the U.S. Automobile Club (USAC) to conduct a survey of the general conditions of the cars rented to the public by Hertz, Avis, National, and Budget Rent-a-Car. USAC officials evaluate each company's cars using a demerit point system. Each car starts with a perfect score of 0 points and incurs demerit points for each discrepancy noted by the inspectors. One measure of the overall condition of a company's cars is the mean of all scores received by the company (i.e. the company's fleet mean score). To estimate the fleet mean score of each rental car company, 10 major airports were randomly selected, and 10 cars from each company were rented for inspection from each airport by USAC officials (i.e. a sample size of 100 cars from each company's fleet was drawn and inspected). Assume that a company's fleet mean score is 30 and standard deviation of their fleet scores is 60. What is the probability that USAC will find an average score of 45 or more for the cars sampled from this company?
- Less than 1%
 - Between 1% and 15%
 - Between 15% and 45%
 - More than 45%
10. While constructing a confidence interval, the margin of error is found to be 6 units. If the margin of error should be kept no greater than 3 units,
- the confidence level should be doubled.
 - the standard deviation of the population should be doubled.
 - the sample size should be doubled.
 - the sample size should be increased by 4 times.

Part II: Show Your Work (3 Problems, 20 Points Total)

Problem 1

The prices of personal computers manufactured by a company based in Dallas are independent and normally distributed with a mean of \$1,050 and a standard deviation of \$200.

- a. Computers with prices of more than \$1,350 receive a discount. What percentage of the computers will receive the discount? (2 pts)

- b. A sample of 100 computers is selected at random. Let $\bar{p}$ be the sample proportion in these 100 computers that would be sold on discount. Calculate the expected value and the standard deviation of $\bar{p}$. (2 pts)

- c. Based on the distribution of $\bar{p}$ from b, what is the probability that, out of these 100 computers, the total number of computers that would be sold on discount is no more than 10? (2 pts)

- d. For a random sample of size 25 of computers, specify the sampling distribution of the sample mean price. What is the probability that the sample mean price of 25 computers is between \$1,000 and \$1,120? (3 pt)

Problem 2

You are considering investing on the stock market. You are thinking of two indices: the S&P 500 and an Emerging Markets Index (EMI). Recent historical data shows that the annual return rate S&P500 follows a normal distribution with expectation of 10% and standard deviation of 15%. The annual return rate of EMI also follows a normal distribution, but with expectation of 20% and standard deviation of 18%. The correlation coefficient of the two indices is 0.2.

You decide to diversify your investment by considering the following two portfolios.

Portfolio A: Invest 80% of your money on S&P500 and the remaining 20% on EMI.

Portfolio B: Invest 60% of your money on S&P500 and the remaining 40% on EMI.

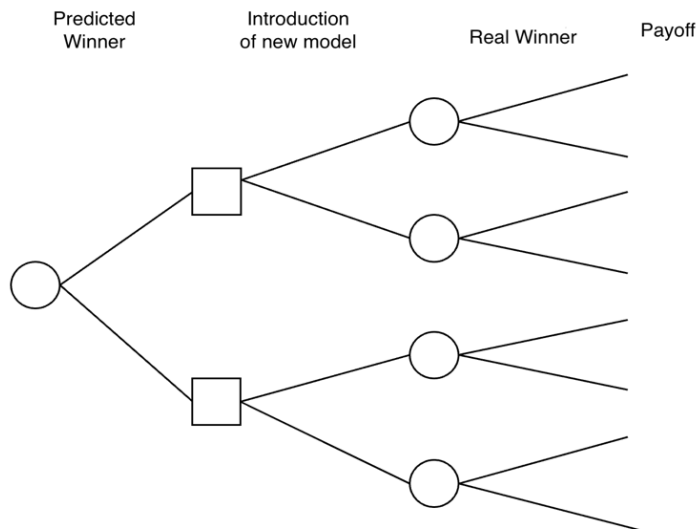
Calculate the expected return and standard deviation of each of the portfolio. Which one is a better investment? (5 pts)

Problem 3

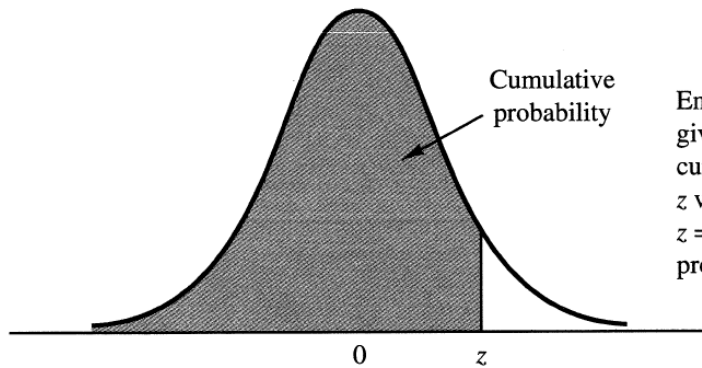
Susan, the marketing manager of a consumer electronics company, is considering whether to introduce a new large screen television model into the market. Susan is aware that the success of the new model depends on the economy in the coming year which in turn depends on the result of the upcoming presidential election. According to her assessment, the incumbent and the challenger are equally likely to win the elections. A fixed cost of 3 million dollars has been incurred prior to making the decision to launch the model. The payoff table (in millions) for the two alternative actions is set up as follows (the fixed cost has been included in the calculation):

Introduction of the new model	Presidential Election Outcome	
	Incumbent	Challenger
Yes	-35	45
No	-3	-3

- Find the expected payoffs of the two alternatives. Which is better? (2 pts)
- Bob argued that he could provide a prediction of the election result through a special survey method. After interviewing Bob, Susan obtained the following probability estimates:
 $P(\text{Predicted Winner is Incumbent} \mid \text{Real Winner is Incumbent}) = .8$
 $P(\text{Predicted Winner is Challenger} \mid \text{Real Winner is Incumbent}) = .2$
 $P(\text{Predicted Winner is Incumbent} \mid \text{Real Winner is Challenger}) = .3$
 $P(\text{Predicted Winner is Challenger} \mid \text{Real Winner is Challenger}) = .7$
 Complete the following decision tree for evaluating the expected payoff with sample information provided by Bob's survey. Compute the expected value of the sample information. (4 pts)



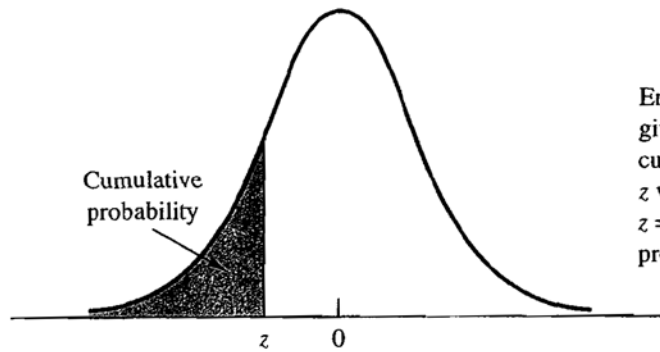
CUMULATIVE PROBABILITIES FOR THE STANDARD NORMAL DISTRIBUTION



Entries in the table give the area under the curve to the left of the z value. For example, for $z = 1.25$, the cumulative probability is .8944.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9913
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9986	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990

CUMULATIVE PROBABILITIES FOR THE STANDARD NORMAL DISTRIBUTION



Entries in this table give the area under the curve to the left of the z value. For example, for $z = -.85$, the cumulative probability is .1977.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

Part I Solutions (Version A)

1. C
2. B Since the distribution is not bell-shaped, we can only apply Chebyshev's theorem which says that at least $1 - \frac{1}{2^2}$ of the restaurants must be within 2 standard deviations of the mean.
3. C The distribution is right-skewed, so average is more than the median. Only 25% of the data is between 2.32% and 4.07%. Note that 5% is above the max value on the box-plot
4. A $X \sim \text{Binomial}(12, 0.1)$. $P(X \geq 1) = 1 - P(X=0) = 1 - 0.2824 = 0.7176$. $P(X \leq 1) = P(X=0) + P(X=1) = 0.2824 + 0.3763 = 0.6587$. So $Q1 > Q2$.
5. D $\frac{101+19}{150} = 0.8$.
6. A Let P be the price of items sold and Z be the error in the price. Then,
 $E[P+Z] = E[P] + E[Z] = E[P]$.
7. B Let C = Set up correctly, I = Set up incorrectly, A = non-defective
 $P(A) = P(A \cap C) + P(A \cap I) = P(A|C)P(C) + P(A|I)P(I)$
 $= .97(.95) + .40(.05) = .9415$
 $P(C|A) = 0.97*0.95 / 0.9415 = 0.9787$.
8. B $Z=1.285$. So $12+3.2*1.285 = 16.1$
9. A $P(\bar{X} \geq 45) = P\left(Z \geq \frac{45-30}{60/\sqrt{100}} = 2.5\right) \approx 0.0062$.
10. D $n_2 = \left(\frac{z_{\alpha/2}\sigma}{\frac{1}{2}E_1}\right)^2 = 4\left(\frac{z_{\alpha/2}\sigma}{E_1}\right)^2 = 4n_1$.

Part II Solutions (Version A)

Problem 1

1.
 - a. $P(X > 1350) = P(Z > 1.5) = 1 - P(Z \leq 1.5) = 1 - 0.9332 = 0.0668$
 - b. $E(\bar{p}) = 0.0668$, $SD(\bar{p}) = 0.025$.
 - c. $P(\text{total number sold on discount} \leq 10) = P(\bar{p} \leq 0.10) = P(Z \leq 1.328) = 0.9082$.

It is also OK to use Binomial distribution to calculate this. Denote Y as the number of computers that would be sold on discount. $Y \sim \text{Binomial}(100, 0.0668)$. Use normal approximation, it is approximately normal with mean $np = 6.68$ and standard deviation $= 2.5$. The Z -value is $(10 - 6.68)/2.5 = 1.33$ and hence $P(Y \leq 10) = 0.9082$. If students use 0.5 continuity factor, the Z -value is $(10.5 - 6.68)/2.5 = 1.53$, and hence $P(Y \leq 10) = 0.937$.

d. $\bar{X}$ is $N\left(1050, \left(\frac{200}{\sqrt{25}}\right)^2\right)$, that is, $\text{Normal}(1050, 40^2)$

$$P(1000 < \bar{X} < 1120) = P(-1.25 < Z < 1.75) = P(Z < 1.75) - P(Z < -1.25) = 0.9599 - 0.1056 = 0.8543$$

Problem 2

Let S denote the return on S&P 500 and M be the return on EMI. Then,

$$E[0.8 \times S + 0.2 \times M] = 0.8 \times E[S] + 0.2 \times E[M] = 12\% .$$

$$E[0.6 \times S + 0.4 \times M] = 0.6 \times E[S] + 0.4 \times E[M] = 14\% .$$

$$\begin{aligned} \text{Var}[0.8 \times S + 0.2 \times M] &= 0.8^2 \times \text{Var}[S] + 0.2^2 \times \text{Var}[M] + 2 \times 0.8 \times 0.2 \times \text{SD}[S] \times \text{SD}[M] \times 0.2 \\ &= 174.24 \end{aligned}$$

$$\text{SD}[A] = \sqrt{174.24} = 13.2\%$$

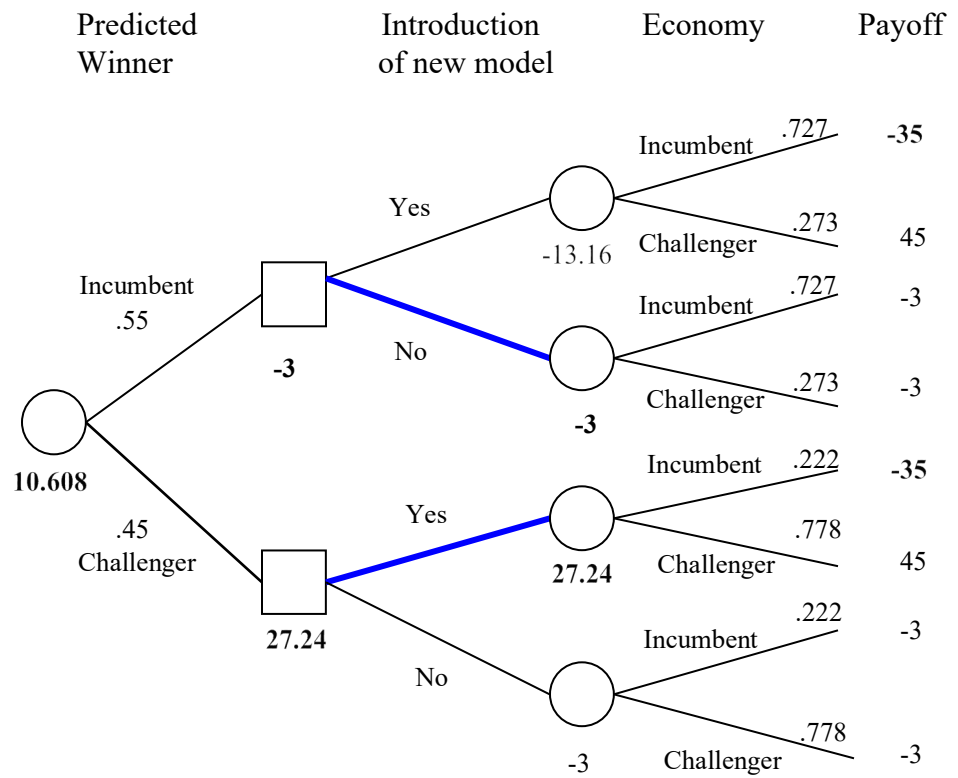
$$\begin{aligned} \text{Var}[0.6 \times S + 0.4 \times M] &= 0.6^2 \times \text{Var}[S] + 0.4^2 \times \text{Var}[M] + 2 \times 0.6 \times 0.4 \times \text{SD}[S] \times \text{SD}[M] \times 0.2 \\ &= 158.76 \end{aligned}$$

$$\text{SD}[B] = \sqrt{158.76} = 12.6\%$$

Investment B is preferable since it has a higher expected return and a lower standard.

Problem 3

- a. EV of introducing the new model = $(-35)(.5) + (45)(.5) = 5$
 EV of not introducing the new model = $(-3)(.5) + (-3)(.5) = -3$
 Therefore, introducing the new model gives a higher expected payoff.
- b. The complete tree is shown in next page.
 $P(\text{Predicted Winner} = \text{Incumbent}) = P(\text{Predicted Winner} = \text{Incumbent} \mid \text{Real Winner} = \text{Incumbent}) \times P(\text{Real Winner} = \text{Incumbent}) + P(\text{Predicted Winner} = \text{Incumbent} \mid \text{Real Winner} = \text{Challenger}) \times P(\text{Real Winner} = \text{Challenger}) = 0.8 \times 0.5 + 0.3 \times 0.5 = 0.55$.
 $P(\text{Predicted Winner} = \text{Challenger}) = 1 - 0.55 = 0.45$.
 $P(\text{Real Winner} = \text{Incumbent} \mid \text{Predicted Winner} = \text{Incumbent}) = P(\text{Predicted Winner} = \text{Incumbent} \mid \text{Real Winner} = \text{Incumbent}) \times P(\text{Real Winner} = \text{Incumbent}) / P(\text{Predicted Winner} = \text{Incumbent}) = 0.8 \times 0.5 / 0.55 = 0.727$
 $P(\text{Real Winner} = \text{Challenger} \mid \text{Predicted Winner} = \text{Incumbent}) = 1 - 0.727 = 0.273$
 $P(\text{Real Winner} = \text{Challenger} \mid \text{Predicted Winner} = \text{Challenger}) = 0.7 \times 0.5 / 0.45 = 0.778$
 $P(\text{Real Winner} = \text{Incumbent} \mid \text{Predicted Winner} = \text{Challenger}) = 1 - 0.778 = 0.222$.



EV with sample information = $0.55*(-3) + 27.24*0.45 = 10.608$
 So EVSI = $10.608 - 5 = 5.608$.